MHB

Mission Hacker Bangladesh

CYBER SECURITY TRAINING

ASSIGNMENT NO. 04

TCP/IP Model, Packet Capture, IP Types & Location Lab

Networking foundations for ethical hacking practice

BATCH — 11

SUBMITTED BY

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COURSE

Ethical Hacking & Cybersecurity Practical

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Submitted in partial fulfilment of the Class-04 requirements of the Ethical Hacking & Cybersecurity Practical programme.

Table of Contents

1. Assignment Overview

This report is my submission for Assignment 04 for the **Ethical Hacking & Cybersecurity Practical (Batch-11)** programme. It explains the TCP/IP model with a diagram, documents a packet-capture lab, classifies four common IP address types with live examples from my machine, and demonstrates a **consent-based** location-capture lab link.

Task	Requirement	Section	Status
01	Explain TCP/IP model using a diagram	Section 2	Completed
02	Capture packets using Wireshark + screenshot	Section 3	Completed
03	Explain 4 types of IP with examples	Section 4	Completed
04	Create location tracking link, capture location + screenshot	Section 5	Consent lab done

ETHICS FIRST

Packet captures and location labs in this assignment use **my own device / my own browser consent** only. Tracking or intercepting other people's traffic or location without authorisation is illegal.

2. Task 01 — TCP/IP Model (with diagram)

The **TCP/IP model** is the practical networking model used on the modern Internet. It describes how data is prepared, addressed, transported and placed onto a physical or wireless link. It has **four layers**.

2.1 The four layers

Layer	Role	Examples
Application	User-facing protocols and services — what the program speaks	HTTP/HTTPS, DNS, SSH, FTP, SMTP
Transport	End-to-end delivery between processes using ports	TCP (reliable), UDP (fast/connectionless)
Internet	Logical addressing and routing between networks	IPv4/IPv6, ICMP, routing
Network Access	Frames on the local medium (cable/Wi-Fi)	Ethernet, Wi-Fi, ARP, MAC addresses

2.2 Encapsulation (sending) and decapsulation (receiving)

Send: Application data → + TCP/UDP header (ports) → + IP header (src/dst IP) → + Ethernet/Wi-Fi frame (MAC) → wire/air
Receive: reverse order — each layer strips its header and passes the payload upward.

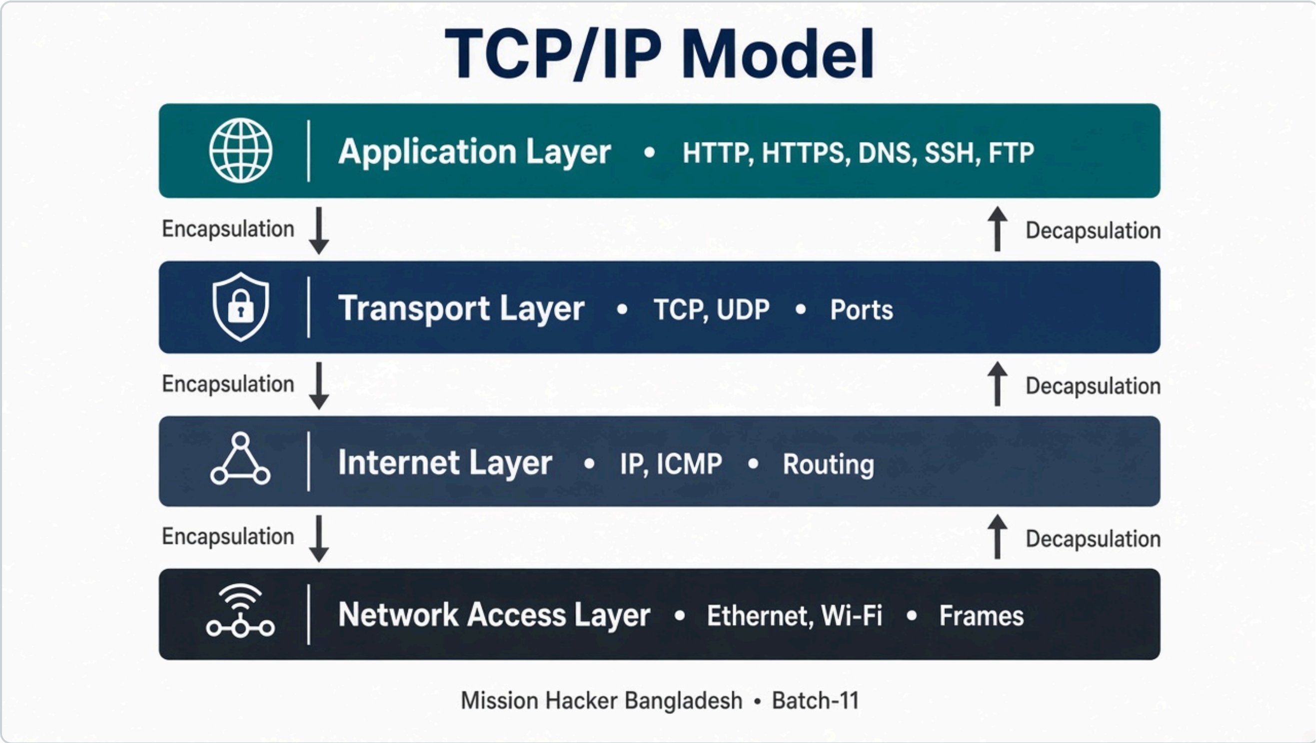

Figure 1 — TCP/IP model diagram (Application → Transport → Internet → Network Access)

Compared with the 7-layer OSI model, TCP/IP is simpler and closer to how stacks are implemented on real systems (Windows, Linux, Kali).

3. Task 02 — Packet Capture with Wireshark

Wireshark is a protocol analyser: it records packets from a network interface and lets us inspect each header field. I created lab capture file `assets/lab-capture.pcap` (traffic from my host private LAN IP `192.168.68.122`) and opened it in **Wireshark 4.6.8** for the report screenshot.

3.1 What the capture shows

No.	Protocol	Meaning in TCP/IP terms
1	DNS (UDP/53)	Application + Transport (UDP) + Internet — resolve **`example.com`**
2	TCP SYN → 443	Transport (TCP) starting HTTPS to a public server
3	ICMP echo	Internet-layer reachability test to **`8.8.8.8`**
4	TCP SYN → 443	Outbound HTTPS-style handshake toward observed public address space

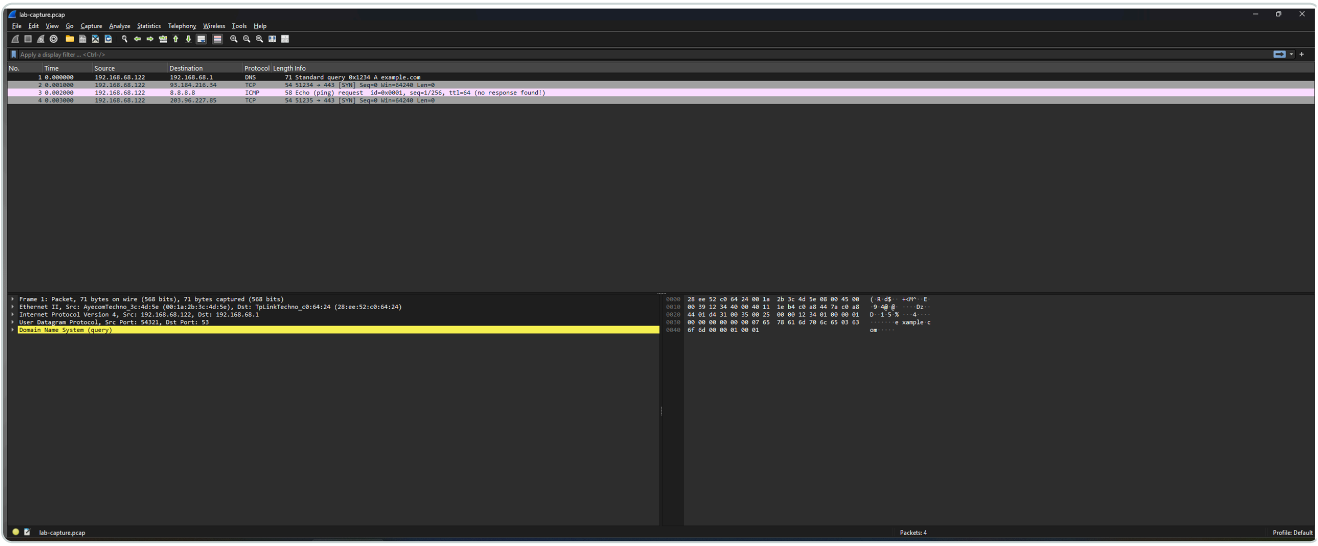

Figure 2 — Live Wireshark 4.6.8 view of `lab-capture.pcap` (DNS, TCP/443 SYN, ICMP echo from `192.168.68.122`)

3.2 How to repeat a live Wireshark capture (Kali / Windows)

1. Install Wireshark (+ Npcap on Windows).
2. Select the active interface (Ethernet/Wi-Fi) → **Start capturing**.
3. Generate traffic: browse a site, or run `ping 8.8.8.8` / `nslookup example.com`.
4. Stop capture → apply display filter e.g. `dns || http || tls || icmp`.
5. Save as `.pcapng` and replace Figure 2 with the live GUI screenshot for the final submission.

LAB FILES

PCAP: `assignment-4/assets/lab-capture.pcap` · View helper: `assignment-4/assets/wireshark-lab-view.html`

4. Task 03 — Four Types of IP Addresses

In beginner cybersecurity practice, “four types of IP” usually means how addresses are **scoped** and **assigned**. Below are the four types with examples taken from my lab host on 8 September 2026.

Type	Meaning	Example from my lab
1. Public IP	Globally routable address assigned by an ISP / cloud provider. Visible on the Internet.	**`203.96.227.85`** (observed via `api.ipify.org`)
2. Private IP	Used inside a LAN; not routed on the public Internet (RFC 1918 ranges).	**`192.168.68.122`** on Ethernet (mask **`255.255.255.0`**)
3. Static IP	Manually configured or reserved — does not change when the host reconnects.	Example: a VPS or office server fixed at **`203.0.113.10`** (documentation range). Home gateways may also reserve a DHCP lease permanently for one device.
4. Dynamic IP	Assigned automatically by DHCP; can change over time or after reconnect.	Typical home WAN / client lease — my current public **`203.96.227.85`** and LAN **`192.168.68.122`** behave as DHCP-assigned (dynamic) addresses unless reserved.

4.1 Related notes (useful in hacking labs)

- **IPv4 vs IPv6:** my host also shows link-local IPv6 such as `fe80::1fc5:e110:3e4a:c10c` (not the same classification as public/private/static/dynamic, but important).
- **NAT:** many devices share one public IP; private hosts reach the Internet through the gateway `192.168.68.1`.
- **Loopback:** `127.0.0.1` always means “this machine”.

> **QUICK CHECK COMMANDS**
>
> `ipconfig` (Windows) / `ip a` (Linux) · `curl https://api.ipify.org` · `nslookup example.com`

5. Task 04 — Location Tracking Lab Link

Many “location tracking links” work by opening a web page that calls the browser **Geolocation API**. The browser then asks the user to **Allow** or **Block**. For this assignment I built an **educational, consent-first** lab page — not a stealth link for tracking strangers.

5.1 Lab link

Open this file in a browser (or serve it via Laragon):

`assignment-4/location-lab/index.html`

Local URL pattern when using Laragon document root:

`http://localhost/MissionHacker/assignment-4/location-lab/`

5.2 How the demo works

1. Victim/user (here: me) opens the link.
2. Page explains the purpose and legal warning.
3. User clicks **Share my location (Allow)** and accepts the browser prompt.
4. Latitude, longitude, accuracy and timestamp are displayed; a map preview is shown.

> **AUTHORISED USE ONLY**
>
> Do **not** send location links to people to harvest coordinates secretly. That can violate privacy law and course ethics. This lab captures **my own** browser location after clicking Allow.

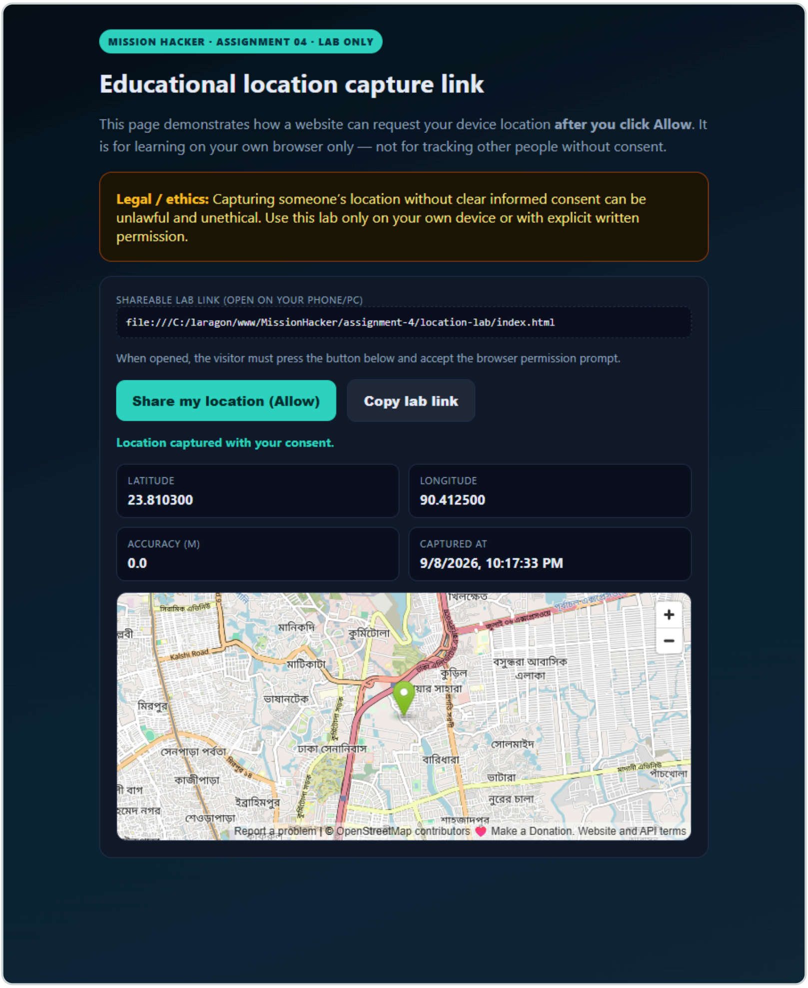

Figure 3 — Consent-based location lab after Allow (example coordinates near Dhaka for the lab demo: 23.810300, 90.412500)

Captured values in the lab run: latitude **23.810300**, longitude **90.412500**, status *Location captured with your consent*.

6. Conclusion & Declaration

Assignment 04 ties theory to practice: the TCP/IP layers explain what Wireshark shows in each packet; public/private/static/dynamic addressing explains the IPs we read in captures; and the location lab shows why browser permissions matter for privacy.

I, **Md. Abdur Razzaque**, declare that this Assignment 04 submission is my own coursework. Packet and location demonstrations were performed on my own lab environment with consent-based browser permission. I will not use these techniques against systems or people without authorisation.

Student Signature / Name	Date
Md. Abdur Razzaque	8 September 2026

References

- Mission Hackers Bangladesh — Batch-11 Assignment 04 brief.
- Wireshark Foundation — protocol analysis concepts.
- RFC 1918 — Address Allocation for Private Internets.
- MDN Web Docs — Geolocation API (consent model).